PEMROGRAMAN BERBASIS FRAMEWORK
JOBSHEET 5

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PROGRAM STUDI TEKNIK INFORMATIKA
JURUSAN TEKNOLOGI INFORMASI
POLITEKNIK NEGERI MALANG
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Praktikum

1. Global CSS

- File global.css

JS_5 > my-app > src > styles > # globals.css > ...

```
* {
  box-sizing: border-box;
  padding: 0;
  margin: 0;
}

html,
body {
  max-width: 100vw;
  overflow-x: hidden;
}

a {
  color: inherit;
  text-decoration: none;
}

.navbar {
  width: 100%;
  height: 60px;
  background-color: #333;
  color: #fff;
  display: flex;
  align-items: center;
  padding: 0 20px;
}
```

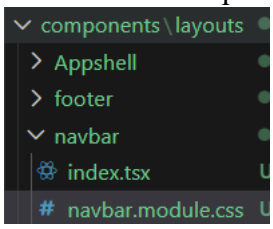
- Import global.css

JS_5 > my-app > src > pages > _app.tsx > ...

```
import '@/styles/globals.css'
import type { AppProps } from 'next/app'
import AppShell from '@/components/layouts/Appshell'
import Navbar from '@/components/layouts/navbar'
```

2. CSS Module (Local Scope)

- Struktur komponen Navbar



- Modifikasi global.css dan navbar.module.css

JS_5 > my-app > src > styles > # globals.css >

```
* {
  box-sizing: border-box;
  padding: 0;
  margin: 0;
}

html,
body {
  max-width: 100vw;
  overflow-x: hidden;
}

a {
  color: inherit;
  text-decoration: none;
}
```

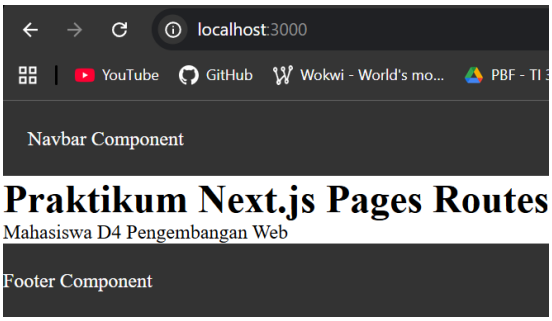
JS_5 > my-app > src > components > layouts > navbar > # navbar.module.css >

```
.navbar {
  width: 100%;
  height: 60px;
  background-color: #333;
  color: #fff;
  display: flex;
  align-items: center;
  padding: 0 20px;
}
```

- Panggil module navbar lalu jalankan

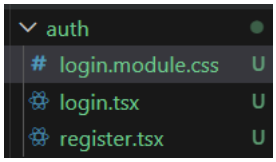
JS_5 > my-app > src > components > layouts > navbar > index.tsx >

```
import style from './navbar.module.css';

const Navbar = () => {
    return (
        <div className={style.navbar}>
            Navbar Component
        </div>
    );
};

export default Navbar;
```



3. Styling untuk Pages (CSS Module)

- Tambahkan file login.module.css pada folder auth



- Modifikasi login.module.css dan login.tsx

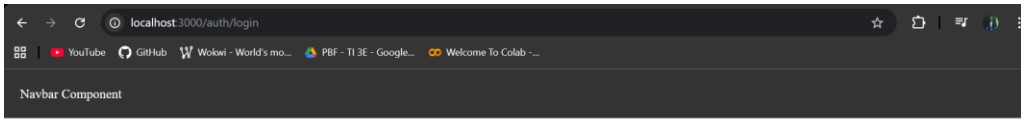
JS_5 > my-app > src > pages > auth > login.tsx > HalamanLogin

```
import Link from "next/link";
import { useRouter } from "next/router";
import style from "./login.module.css";

const HalamanLogin = () => {
    const router = useRouter();
    const handleLogin = () => {
        // Login logic here
        router.push("/produk");
    };
    return (
        <div className={style.login}>
            <h1>Halaman Login</h1>
            <button onClick={handleLogin}>Login</button>

            <p>
                Belum punya akun?
                <Link href="/auth/register">
                    Ke Halaman Register
                </Link>
            </p>
        </div>
    );
};

export default HalamanLogin;
```

JS_5 > my-app > src > pages > auth > # login.module.css >

```
.login{
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    height: 100vh;
}
```

- Jalankan browser



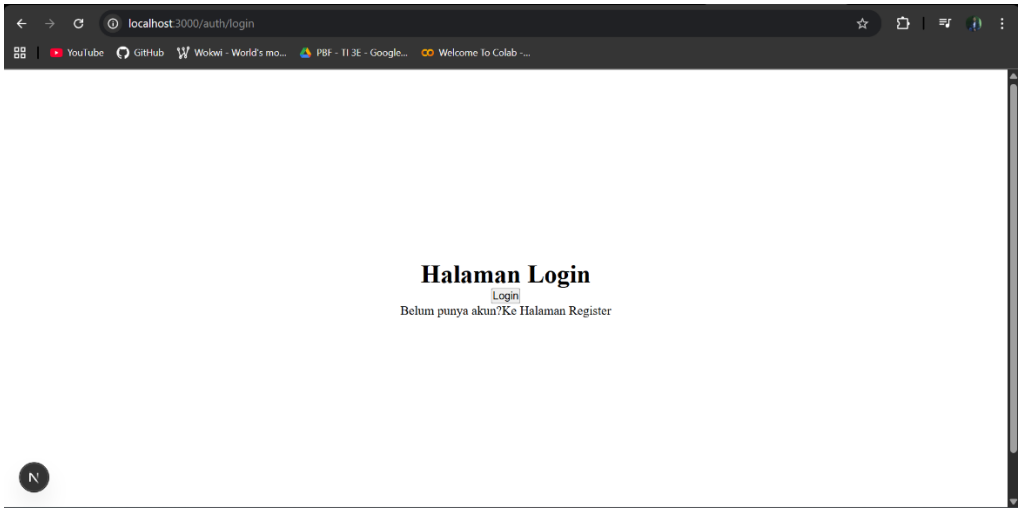
4. Conditional Rendering Navbar (Tanpa Navbar di Login)

- Modifikasi index.tsx pada folder Appsheel

JS_5 > my-app > src > components > layouts > Appshell > index.tsx > AppShell

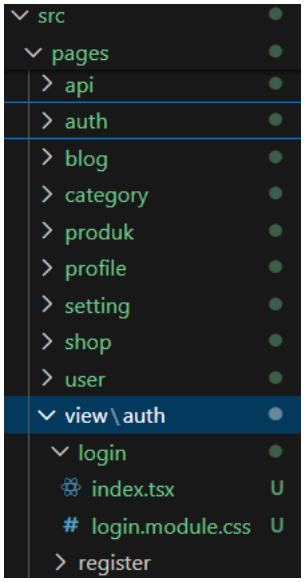
```tsx
import { useRouter } from "next/router";
import Navbar from "../navbar";
import Footer from "../footer";

const disableNavbar = ["/auth/login", "/auth/register"];

type AppShellProps = {
    children: React.ReactNode;
};

const AppShell = (props: AppShellProps) => {
    const { children } = props;
    const { pathname } = useRouter();
    return (
        <main>
            {!disableNavbar.includes(pathname) && <Navbar />}
            {children}
            <Footer />
        </main>
    );
};

export default AppShell;
```

- Jalankan browser



5. Refactoring Struktur Project (Best Practice)

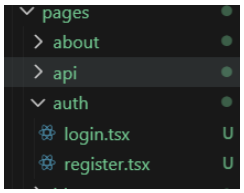
- Refactor struktur login



- Modifikasi login.module.css pada folder view/auth/login/ dan hapus yang ada pada pages/auth

JS_5 > my-app > src > pages > view > auth > login > # login.module.css >

```
.login{
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    height: 100vh;
}
```



- Modifikasi login.tsx pada folder pages/auth

JS_5 > my-app > src > pages > auth > login.tsx > default

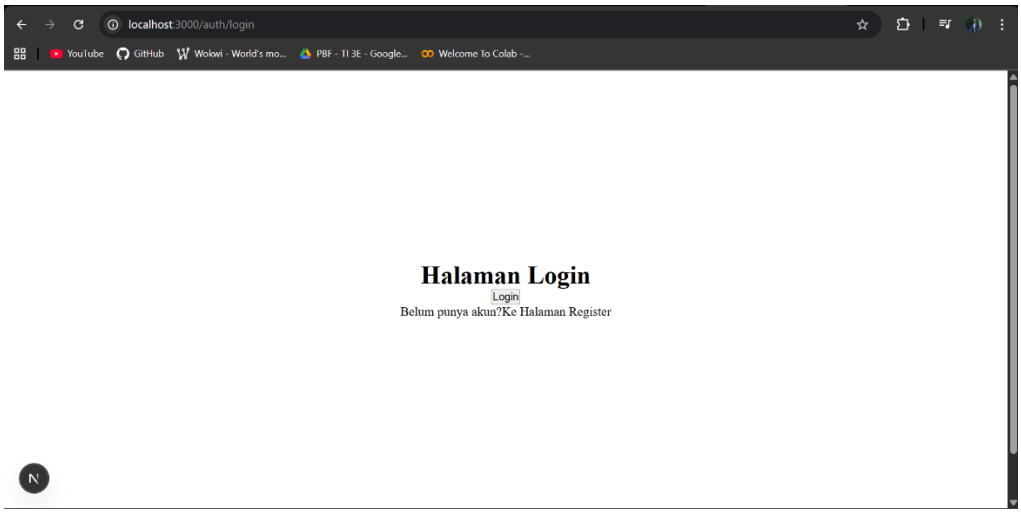
```
import TampilanLogin from "../view/auth/login";

const HalamanLogin = () => {
    return (
        <div>
            <TampilanLogin />
        </div>
    );
};

export default HalamanLogin;
```

- Modifikasi index.tsx pada folder views/auth/login

JS_5 > my-app > src > pages > view > auth > login > index.tsx > TampilanLogin

```
import Link from "next/link";
import { useRouter } from "next/router";
import style from "./login.module.css";

const TampilanLogin = () => {
    const router = useRouter();
    const handleLogin = () => {
        // Login logic here
        router.push("/produk");
    };
    return (
        <div className={style.login}>
            <h1>Halaman Login</h1>
            <button onClick={handleLogin}>Login</button>

            <p>
                Belum punya akun?
                <Link href="/auth/register">
                    Ke Halaman Register
                </Link>
            </p>
        </div>
    );
};

export default TampilanLogin;
```

- Jalankan browser



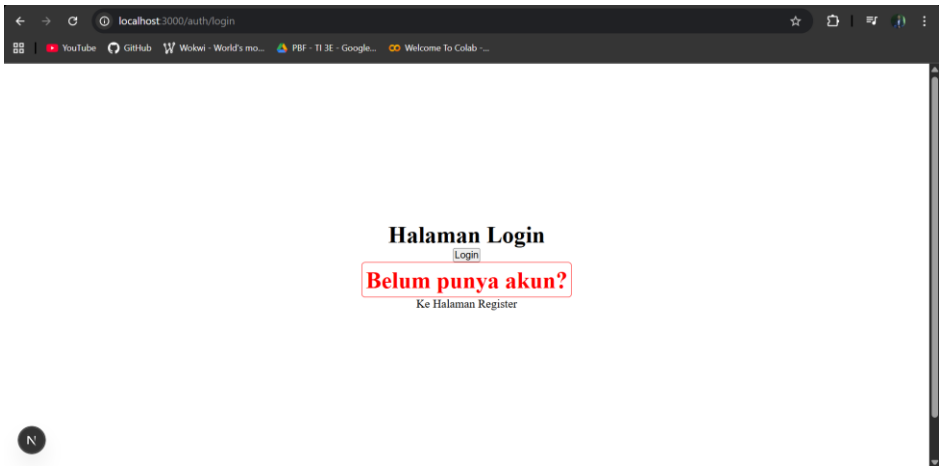
6. Inline Styling (CSS-in-JS)

- Modifikasi index.tsx pada folder views/auth/login

JS_5 > my-app > src > pages > view > auth > login > index.tsx > TampilanLogin

```
import Link from "next/link";
import { useRouter } from "next/router";
import style from "./login.module.css";

const TampilanLogin = () => {
    const router = useRouter();
    const handleLogin = () => {
        // Login logic here
        router.push("/produk");
    };
    return (
        <div className={style.login}>
            <h1>Halaman Login</h1>
            <button onClick={handleLogin}>Login</button>
            <h1 style={{color:'red',border:'1px solid red',borderRadius:'5px',padding:'5px'}}>Belum punya akun?</h1>
            <Link href="/auth/register">Ke Halaman Register</Link>
        </div>
    );
};

export default TampilanLogin;
```

- Jalankan browser



7. Kombinasi Global CSS + CSS Module

- Modifikasi global.css

JS_5 > my-app > src > styles > # globals.css >

```
* {
    box-sizing: border-box;
    padding: 0;
    margin: 0;
}

html,
body {
    max-width: 100vw;
    overflow-x: hidden;
}

a {
    color: inherit;
    text-decoration: none;
}

.big {
    font-size: 1.5rem;
}
```

- Modifikasi index.tsx pada folder components/layouts/navbar

JS_5 > my-app > src > components > layouts > navbar > index.tsx

```tsx
import style from './navbar.module.css';

const Navbar = () => {
    return (
        <div className={style.navbar}>
            <div className='big'>
                Navbar Component
            </div>
        </div>
    );
};

export default Navbar;
```

8. SCSS (SASS)

- Install SASS lalu cek pada package.css

```
PS E:\PBF_2026\JS_5\my-app> npm i --save-dev sass
```

JS_5 > my-app > package.json > ...

```jsx
{
  "name": "my-app",
  "version": "0.1.0",
  "private": true,
  ▷Debug
  "scripts": {
    "dev": "next dev",
    "build": "next build",
    "start": "next start",
    "lint": "next lint"
  },
  "dependencies": {
    "@types/node": "25.2.3",
    "@types/react": "19.2.14",
    "@types/react-dom": "19.2.3",
    "eslint": "9.39.2",
    "eslint-config-next": "16.1.6",
    "next": "16.1.6",
    "react": "19.2.4",
    "react-dom": "19.2.4",
    "typescript": "5.9.3"
  },
  "devDependencies": {
    "sass": "^1.97.3"
  }
}
```

- Tambahkan colors.scss pada folder styles dan Modifikasi colors.scss

```
styles
  colors.scss        U
  globals.css        U
  Home.module.css    U
```

JS_5 > my-app > src > styles > colors.scss

```scss
$schema:(
    color-primary: #3498db,
    color-secondary: #2ecc71,
    color-accent: #e74c3c,
    color-background: #ecf0f1,
    color-text: #2c3e50
)
```

- Tambahkan file login.module.scss pada folder views/auth/login/

```
view\auth
  login
    index.tsx          U
    login.module.css   U
    login.module.scss  U
```

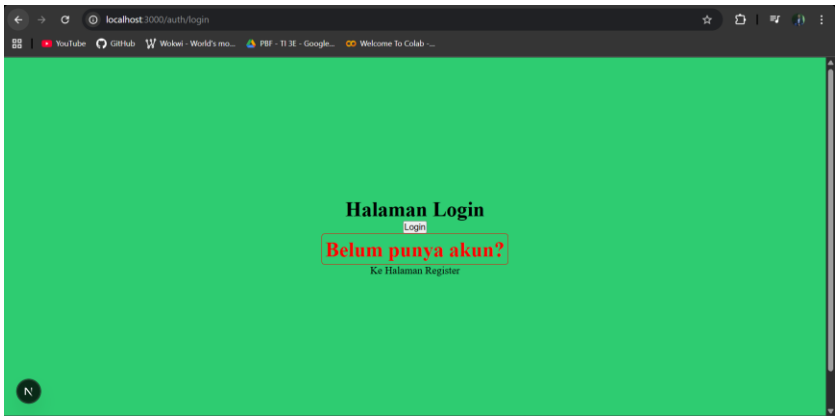
- Modifikasi index.tsx

```
JS_5 > my-app > src > pages > view > auth > login > index.tsx > default
import Link from "next/link";
import { useRouter } from "next/router";
//import style from "./login.module.css";
import style from "./login.module.scss";

const TampilanLogin = () => {
    const router = useRouter();
    const handleLogin = () => {
        // Login logic here
        router.push("/produk");
    };
    return (
        <div className={style.login}>
            <h1>Halaman Login</h1>
            <button onClick={handleLogin}>Login</button>
            <h1 style={{color:'red',border:'1px solid red',borderRadius:'5px',padding:'5px'}}>Belum punya akun?</h1>
            <Link href="/auth/register">Ke Halaman Register</Link>
        </div>
    );
};

export default TampilanLogin;
```

- Modifikasi login.module.scss

```
JS_5 > my-app > src > pages > view > auth > login > login.module.scss > .login
@import "../../styles/colors.scss";

.login{
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    height: 100vh;
    background-color: map-get($map: $schema, $key:color-secondary);
}
```

- Jalankan browser



9. Tailwind CSS

- Install tailwind dengan printah npm install -D tailwindcss postcss autoprefixer

```
PS E:\PBF_2026\JS_5\my-app> npm install -D tailwindcss postcss autoprefixer

added 5 packages, changed 2 packages, and audited 354 packages in 9s

145 packages are looking for funding
  run `npm fund` for details
```

- Jalankan perintah npx tailwindcss init -p, jika error solusinya downgrade versi tailwindcss nya

```
PS E:\PBF_2026\JS_5\my-app> npx tailwindcss init -p
npm error could not determine executable to run
npm error A complete log of this run can be found in: C:\Users\Fahreza\AppData\Local\npm-cache\_logs\2026-03-03T15_33_08_273Z-debug-0.log
PS E:\PBF_2026\JS_5\my-app> npm install -D tailwindcss@3 postcss autoprefixer

added 35 packages, changed 1 package, and audited 389 packages in 9s

152 packages are looking for funding
  run `npm fund` for details
```

```
PS E:\PBF_2026\JS_5\my-app> npx tailwindcss init -p

Created Tailwind CSS config file: tailwind.config.js
Created PostCSS config file: postcss.config.js
```

- Konfigurasi tailwind.config.js

```
JS_5 > my-app > JS tailwind.config.js > <unknown> > conte
/** @type {import('tailwindcss').Config} */
module.exports = {
  content: [
    "./app/**/*.{js,ts,jsx,tsx,mdx}",
    "./pages/**/*.{js,ts,jsx,tsx,mdx}",
    "./components/**/*.{js,ts,jsx,tsx,mdx}",
    "./src/**/*.{js,ts,jsx,tsx,mdx}",
  ],
  theme: {
    extend: {},
  },
  plugins: [],
}
```

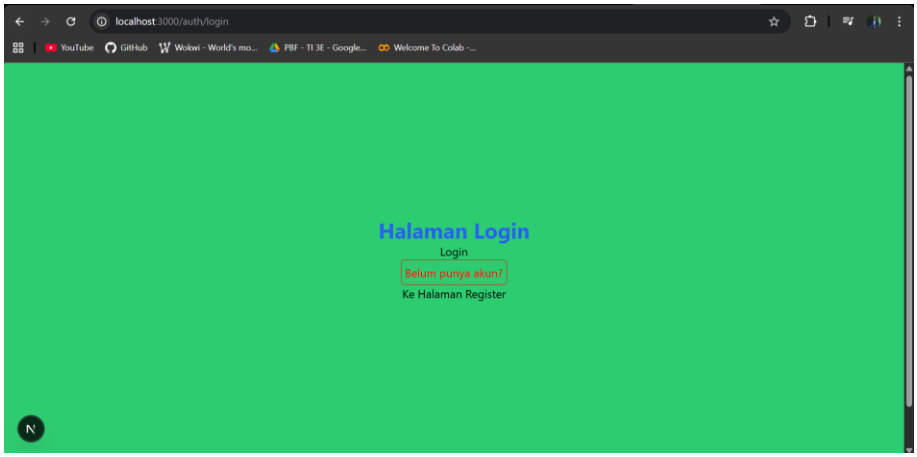
- Import di Global CSS

```
JS_5 > my-app > src > styles > # globals.css >
@tailwind base;
@tailwind components;
@tailwind utilities;
```

- Modifikasi index.tsx pada folder auth/login/

```
JS_5 > my-app > src > pages > view > auth > login > index.tsx > TampilanLogin
import Link from "next/link";
import { useRouter } from "next/router";
//import style from "./login.module.css";
import style from "./login.module.scss";

const TampilanLogin = () => {
    const router = useRouter();
    const handleLogin = () => {
        // Login logic here
        router.push("/produk");
    };
    return (
        <div className={style.login}>
            <h1 className="text-3xl font-bold text-blue-600">Halaman Login</h1>
            <button onClick={handleLogin}>Login</button>
            <h1 style={{color:'red',border:'1px solid red',borderRadius:'5px',padding:'5px'}}>Belum punya akun?</h1>
            <Link href="/auth/register">Ke Halaman Register</Link>
        </div>
    );
};

export default TampilanLogin;
```

- Jalankan browser



Tugas Praktikum

Tugas 1

- Buat halaman Register
- Gunakan CSS Module
  - pages\auth\register.tsx

JS_5 > my-app > src > pages > auth > register.tsx > default

```python
import TampilanRegister from "../view/auth/register";

const HalamanRegister = () => {
    return (
        <div>
            <TampilanRegister />
        </div>
    );
};

export default HalamanRegister;
```

  - pages\view\auth\register\index.tsx

JS_5 > my-app > src > pages > view > auth > register > index.tsx > default

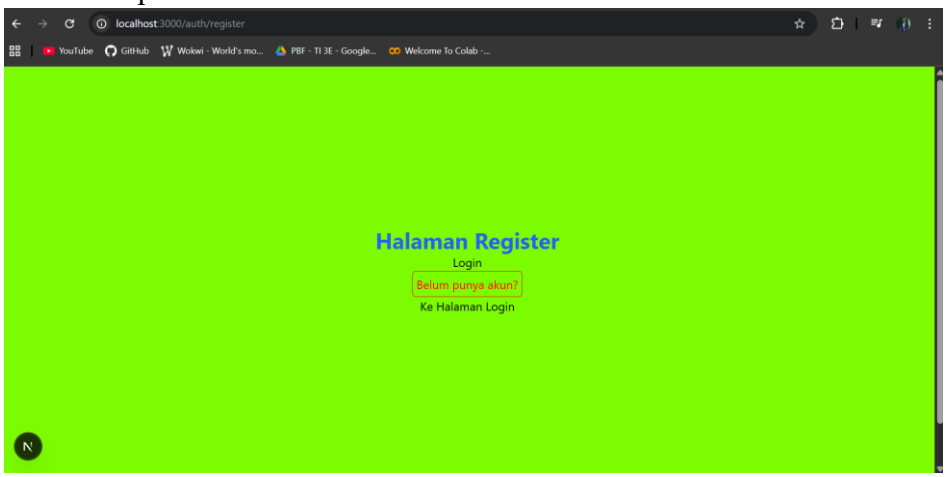
```python
import Link from "next/link";
import { useRouter } from "next/router";
import style from "./register.module.css";

const TampilanRegister = () => {
    const router = useRouter();
    const handleLogin = () => {
        // Login logic here
        router.push("/produk");
    };
    return (
        <div className={style.register}>
            <h1 className="text-3xl font-bold text-blue-600">Halaman Register</h1>
            <button onClick={handleLogin}>Login</button>
            <h1 style={{color:'red',border:'1px solid red',borderRadius:'5px',padding:'5px'}}>Belum punya akun?</h1>
            <Link href="/auth/login">Ke Halaman Login</Link>
        </div>
    );
};

export default TampilanRegister;
```

  - pages\view\auth\register\register.module.css

JS_5 > my-app > src > pages > view > auth > register > register.module.css >

```python
.register {
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    height: 100vh;
    background-color: lawngreen;
}
```

  - Output



Tugas 2

- Refactor halaman Produk ke folder views
- Pisahkan Hero Section dan Main Section
  - HeroSection.tsx

JS_5 > my-app > src > pages > view > produk > HeroSection.

```
import styles from "./produk.module.scss";

const HeroSection = () => {
    return (
        <section className={styles.hero}>
            <h1>Halaman Produk</h1>
            <p>Daftar produk tersedia</p>
        </section>
    );
};

export default HeroSection;
```

  - MainSection.tsx

```
import styles from "./produk.module.scss";

const MainSection = () => {
    return (
        <section className={styles.mainSection}>
            <div className={styles.container}>
                <h2>Daftar Produk</h2>
                <div className={styles.productGrid}>
                    <div className={styles.productCard}>
                        <h3>Produk 1</h3>
                        <p>Deskripsi produk 1</p>
                        <p className={styles.price}>Rp 100.000</p>
                    </div>
                    <div className={styles.productCard}>
                        <h3>Produk 2</h3>
                        <p>Deskripsi produk 2</p>
                        <p className={styles.price}>Rp 200.000</p>
                    </div>
                    <div className={styles.productCard}>
                        <h3>Produk 3</h3>
                        <p>Deskripsi produk 3</p>
                        <p className={styles.price}>Rp 150.000</p>
                    </div>
                    <div className={styles.productCard}>
                        <h3>Produk 4</h3>
                        <p>Deskripsi produk 4</p>
                        <p className={styles.price}>Rp 300.000</p>
                    </div>
                </div>
            </div>
        </section>
    );
};

export default MainSection;
```

- View/produk/Index.tsx

```
JS_5 > my-app > src > pages > view > produk > index.tsx > ...
import HeroSection from "./HeroSection";
import MainSection from "./MainSection";

const TampilanProduk = () => {
    // const [isLogin, setIsLogin] = useSt
    // const { push } = useRouter();

    // useEffect(() => {
    //     if (!isLogin) {
    //         push('/auth/login');
    //     }
    // }, []);

    return (
        <div>
            <HeroSection />
            <MainSection />
        </div>
    );
};

export default TampilanProduk;
```

- View/login/index.tsx

```
import Link from "next/link";
import { useRouter } from "next/router";
//import style from "./login.module.css";
import style from "./login.module.scss";

const TampilanLogin = () => {
    const router = useRouter();
    const handleLogin = () => {
        // Login logic here
        router.push("../produk");
    };
    return (
        <div className={style.login}>
            <h1 className="text-3xl font-bold text-blue-600">Halaman Login</h1>
            <button onClick={handleLogin}>Login</button>
            <h1 style={{color:'red',border:'1px solid red',borderRadius:'5px',padding:'5px'}}>Belum punya akun?</h1>
            <Link href="/auth/register">Ke Halaman Register</Link>
        </div>
    );
};

export default TampilanLogin;
```

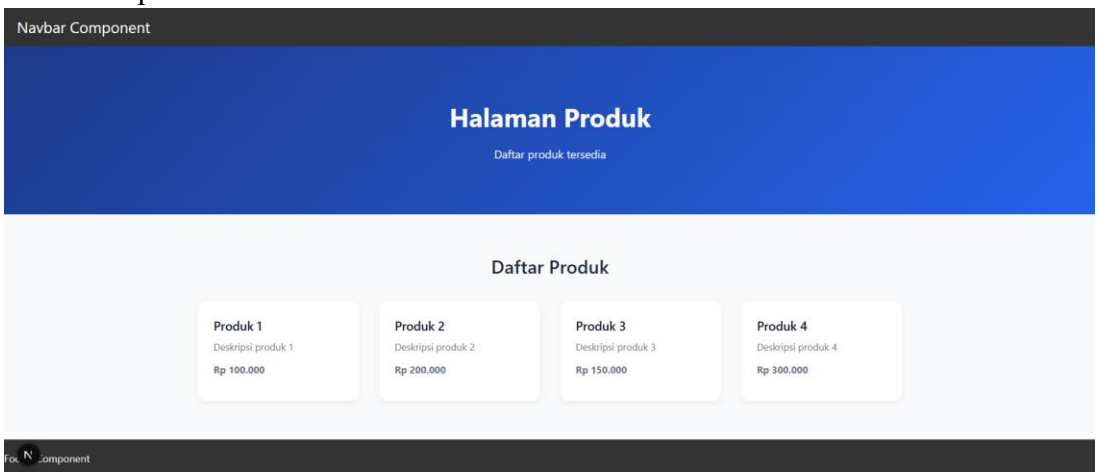
- Produk.module.scss

```
/* HERO SECTION */
.hero {
    background: linear-gradient(135deg, #1e3a8a, #2563eb);
    color: #fff;
    padding: 80px 20px;
    text-align: center;

    h1 {
        font-size: 2.5rem;
        font-weight: 700;
        margin-bottom: 1rem;
    }

    p {
        font-size: 1.1rem;
        opacity: 0.9;
    }
}

/* MAIN SECTION */
.mainSection {
    padding: 60px 20px;
    background-color: #f9fafb;
}

.container {
    max-width: 1100px;
    margin: 0 auto;

    h2 {
        font-size: 1.8rem;
        font-weight: 600;
        margin-bottom: 2rem;
        text-align: center;
        color: #1f2937;
    }
}

/* PRODUCT GRID */
.productGrid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
    gap: 2rem;
}

/* PRODUCT CARD */
.productCard {
    background: #ffffff;
    border-radius: 12px;
    padding: 1.5rem;
    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.05);
    transition: all 0.3s ease;
    cursor: pointer;

    &:hover {
        transform: translateY(-5px);
        box-shadow: 0 10px 20px rgba(0, 0, 0, 0.1);
    }

    h3 {
        font-size: 1.2rem;
        font-weight: 600;
        margin-bottom: 0.5rem;
        color: #111827;
    }

    p {
        color: #6b7280;
        font-size: 0.95rem;
        margin-bottom: 0.8rem;
    }
}

/* PRICE */
.price {
    font-weight: 700;
    font-size: 1rem;
    color: #2563eb;
}
```

- Pages/produk/index.tsx

```
import TampilanProduk from "../view/produk";

const produk = () => {
    /*const [isLogin, setIsLogin] = useState
    const {push} = useRouter();
    useEffect(() => {
        if (!isLogin) {
            push("/auth/login");
        }
    }, []);*/
    return (
        <div>
            <TampilanProduk />
        </div>
    );
};

export default produk;
```

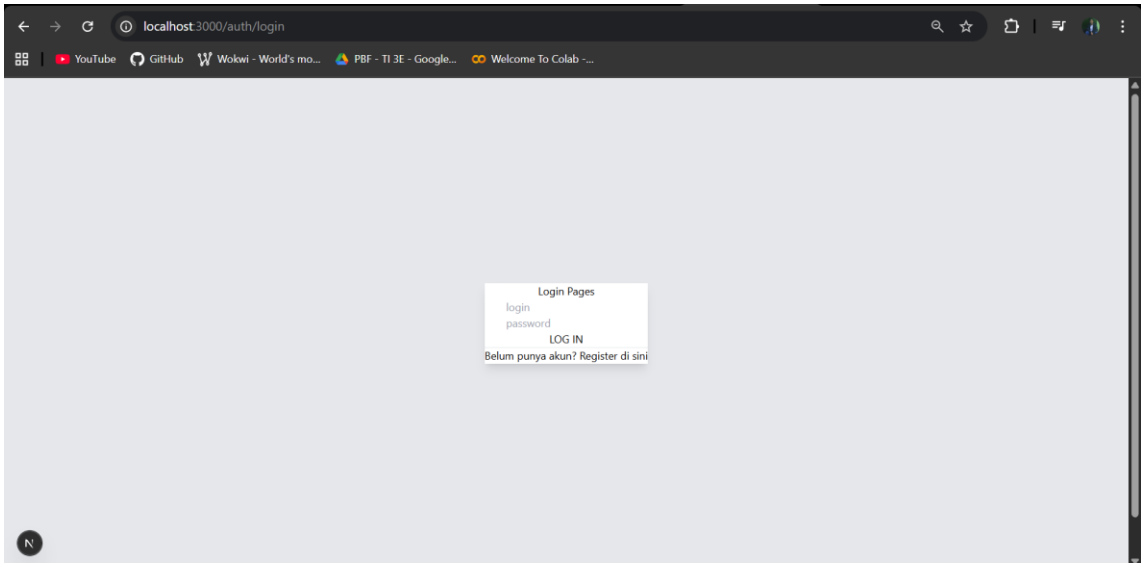
- Output



Tugas 3

- Terapkan Tailwind CSS
- Gunakan minimal 5 utility class
  - Utility Class yang Digunakan: min-h-screen, flex, bg-gray-200, m-auto, shadow-lg

```
JS_5 > my-app > src > pages > view > auth > login > index.tsx > TampilanLogin
import Link from "next/link";
import { useRouter } from "next/router";
//import style from "./login.module.css";
import style from "./login.module.scss";

const TampilanLogin = () => {
    const router = useRouter();
    const handleLogin = () => {
        // Login logic here
        router.push("../produk");
    };
    return (
        <div className="min-h-screen flex bg-gray-200">

            <div className="m-auto shadow-lg bg-white text-center">
                <h3>Login Pages</h3>
                <div>
                    <input type="text" placeholder="login" />
                </div>
                <div>
                    <input type="password" placeholder="password" />
                </div>
                <button onClick={handleLogin}>
                    LOG IN
                </button>
                <hr />
                <p>
                    <Link href="/auth/register">Belum punya akun? Register di sini</Link>
                </p>
            </div>
        </div>
    );
};

export default TampilanLogin;
```



Pertanyaan Refleksi

1. Kapan sebaiknya menggunakan CSS Module dibanding Global CSS?
   Jawab:
   CSS Module sebaiknya digunakan untuk komponen spesifik agar gaya (*style*) tidak bocor atau bertabrakan dengan komponen lain (*local scope*). Sedangkan Global CSS hanya digunakan untuk pengaturan dasar yang berlaku di seluruh halaman, seperti reset margin atau pengaturan font utama
2. Apa kelemahan inline styling?
   Jawab:

- Tidak dapat digunakan kembali (*reusable*): Kode harus ditulis ulang di setiap elemen.
- Keterbatasan fitur: Tidak mendukung *media queries* (untuk responsif) dan *pseudo-classes* (seperti :hover).
- Sulit dikelola: Membuat file JSX/HTML menjadi kotor dan sulit dibaca jika gaya yang diterapkan terlalu banyak.

3. Mengapa SCSS cocok untuk project skala besar?
   Jawab:
   Karena SCSS memiliki fitur Nesting (penulisan kode berjenjang), Variables (untuk menyimpan warna/ukuran standar), dan Mixins/Modules yang memungkinkan pengorganisasian kode CSS menjadi jauh lebih rapi, terstruktur, dan mudah dipelihara saat project bertambah kompleks.
4. Apa keunggulan Tailwind dibanding CSS tradisional?
   Jawab:

- Kecepatan pengembangan: Tidak perlu bolak-balik antara file HTML dan file CSS (cukup gunakan *utility classes*).
- Konsistensi: Ukuran *spacing*, warna, dan tipografi sudah terstandarisasi.
- Ukuran file lebih kecil: Tailwind secara otomatis menghapus CSS yang tidak digunakan (*purging*) saat proses produksi.